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Fearing the Fed: How Wall Street Reads Main Street

Contents

1 Big picture

- **Question.** Why does the stock market sometimes treat good macroeconomic news as good news, but at other times treat the same kind of news as bad news because it implies a more aggressive Federal Reserve?
- **Core object.** The paper estimates the time-varying sensitivity of high-frequency stock futures returns to surprise components of scheduled macroeconomic news announcements (MNAs).
- **Main empirical fact.** Stock market sensitivity to macro news is strongly countercyclical with respect to the output gap: the response is largest when the economy is below trend, especially early in expansions, and close to zero late in expansions.
- **Important novelty.** The strongest variation is *within expansions*, not simply between recessions and expansions. This shifts the interpretation away from recession-volatility stories and toward monetary-policy-expectations stories.
- **Cash-flow versus discount-rate logic.** Positive macro news tends to raise expected cash flows, but it may also raise expected future risk-free rates if investors expect the Fed to tighten. The observed stock response depends on which component dominates.
- **Mechanism.** When the output gap is large and negative, good macro news is less likely to trigger aggressive monetary tightening, so the cash-flow effect is less offset by higher risk-free rates. When the economy is close to peak, good news is more likely to be interpreted as Fed-tightening news.
- **Bond-market evidence.** Bond futures responses around the same MNAs move in the opposite direction to stocks and become stronger when the economy is above trend, consistent with news about future rates offsetting stock cash-flow news.
- **Survey evidence.** Survey-based estimates of perceived policy responsiveness line up with the time variation in stock sensitivity, reinforcing the interpretation that investors' perceived Fed reaction function is changing over time.
- **Post-Covid extension.** During 2022–2023, positive macro surprises sometimes produce negative stock returns because inflation, not just the output gap, becomes a key input into expected monetary tightening.
- **Takeaway.** Wall Street reads Main Street through the Fed. The same macro news has different effects on stock prices depending on how investors think the Fed will respond.

2 Data and measurement

2.1 Macroeconomic news announcements

- The paper uses scheduled U.S. macroeconomic news announcements from Bloomberg Financial Services.
- Bloomberg reports both the realized announcement and a median survey forecast from professional forecasters.
- The baseline sample runs from January 1999 through December 2019; an extension runs through June 2023.
- Announcements include labor-market, production, sentiment, spending, housing, inflation, and other macro releases.
- Most releases are monthly. Initial jobless claims are weekly and GDP growth is quarterly.
- Most announcements occur at 8:30 a.m. or 10:00 a.m. Eastern time, while industrial production is released at 9:15 a.m.

Interpretation: The scheduled nature of MNAs makes them useful for high-frequency identification. In a narrow window around a pre-scheduled release, the surprise component of the announcement is the main new information entering prices.

2.2 Standardized MNA surprises

For announcement i at time t, the paper defines the raw surprise as realized news minus the forecast:

$$\text{MNA}_{i,t} - E_{t-\Delta}(\text{MNA}_{i,t}). \tag{1}$$

The standardized surprise is

$$X_{i,t} = \frac{\text{MNA}_{i,t} - E_{t-\Delta}(\text{MNA}_{i,t})}{\text{Normalization}}. \tag{2}$$

- The benchmark normalization divides by the cross-sectional standard deviation of individual forecasters' predictions for that announcement.
- This makes a surprise economically interpretable relative to forecaster disagreement.
- A robustness normalization divides by the unconditional time-series standard deviation of each announcement surprise.

Interpretation: Standardization is necessary because announcements are measured in different units. It lets the paper compare sensitivities across payrolls, jobless claims, ISM, CPI, GDP, and other releases.

2.3 Financial market data

- The benchmark equity return is from S&P 500 E-mini futures (ES).
- The authors use futures rather than only cash equity prices because many MNAs occur before regular equity market open.
- For bond-market evidence, they use Eurodollar futures and Treasury futures at different maturities.
- Intraday prices come from TickData. The first transaction in each minute is used, with fill-forward if no trade occurs within a minute.
- VIX comes from CBOE and the price-dividend ratio comes from Robert Shiller's data.

Interpretation: Futures markets allow the authors to measure immediate reactions to 8:30 a.m. announcements, which would be missed or contaminated in regular-hours cash equity data.

2.4 Macroeconomic state variables

The paper studies whether the news sensitivity coefficient is explained by state variables $Z_{\tau-1}$, including:

- the output gap, measured as the difference between actual output and potential output;
- changes in the one-year Treasury bill rate;
- the level of Treasury bill rates;
- inflation;
- term premia;
- the price-dividend ratio;
- recession probability or recession dummies;
- VIX and other risk-appetite variables.

Interpretation: The output gap is the central state variable because it captures where the economy is relative to trend. The paper's main claim is that stock responses are largest when the output gap is negative and smallest when the economy is near peak.

3 Models and empirical specifications

3.1 High-frequency return window

For each announcement, the authors compute high-frequency futures returns in a window around the release:

$$r_{t-\Delta_l}^{t+\Delta_h}. \tag{3}$$

The benchmark uses symmetric 30-minute windows before and after the announcement, but the paper examines many combinations of Δ_l and Δ_h.

Interpretation: Varying the window tests whether the measured response is driven by microstructure noise, delayed price discovery, or unrelated market movements.

3.2 Main nonlinear sensitivity regression

The paper estimates, over subperiod τ,

$$r_{t-\Delta_l}^{t+\Delta_h} = \alpha_\tau + \beta_\tau \gamma' X_t + \varepsilon_t. \tag{1}$$

- X_t is the vector of standardized MNA surprises.
- γ measures average responses to the different announcements.
- β_τ is the time-varying sensitivity coefficient.
- $\gamma' X_t$ is a scalar macro-news factor.

Interpretation: The coefficient β_τ says how aggressively stocks respond to the same news factor in period τ. Large β_τ means macro news moves stocks strongly; small β_τ means the same news barely moves stocks.

3.3 Restricted scalar-factor regression

Using the estimated news factor $\widehat{X}_t = \widehat{\gamma}' X_t$, the paper also estimates

$$r_{t-\Delta_l}^{t+\Delta_h} = \alpha_\tau + \beta_\tau \widehat{X}_t + \varepsilon_t. \tag{2}$$

Interpretation: This restricted regression makes it easier to compare return windows and decompositions. The news factor is fixed, so variation comes from the changing scalar sensitivity β_τ.

3.4 Good-news and bad-news decomposition

To ask whether good and bad surprises have different sensitivity, the paper splits the macro-news factor into positive and negative components:

$$r_{t-\Delta}^{t+\Delta} = \alpha_\tau + \beta_\tau^{good} \gamma' X_t^{good} + \beta_\tau^{bad} \gamma' X_t^{bad} + \varepsilon_t. \tag{3}$$

Interpretation: If the time-variation were driven only by reactions to bad news or good news, the two coefficients would move differently. The paper finds broadly similar dynamics, so the main cyclical pattern is not a simple sign-asymmetry effect.

3.5 State-variable restriction on sensitivity

To explain the time variation in β_τ, the paper imposes a parametric restriction:

$$r_{t-\Delta}^{t+\Delta} = \alpha_\tau + \beta_\tau \gamma' X_t + \varepsilon_t, \qquad \beta_\tau = \beta_0 + \beta_1' Z_{\tau-1}. \tag{4}$$

- $Z_{\tau-1}$ contains macro and financial predictors.
- The main predictor is the output gap.
- The post-Covid analysis allows coefficients to differ before and after 2020.

Interpretation: This specification turns the estimated sensitivity series into an economic object. It asks which macro states predict whether stocks will react strongly to news.

3.6 Bond-market version of the state-variable regression

For bond futures, the same state-variable restriction is estimated using bond returns instead of stock returns:

$$r_{t-\Delta}^{bond,t+\Delta} = \alpha_\tau + \beta_\tau^{bond} \gamma' X_t + u_t, \qquad \beta_\tau^{bond} = \beta_0^{bond} + (\beta_1^{bond})' Z_{\tau-1}. \tag{4}$$

Interpretation: If positive macro news raises expected future rates, bond prices should fall. Hence bond betas should often have the opposite sign to stock betas, especially when the Fed is expected to react strongly.

4 Identification and empirical design

4.1 Why high-frequency announcement windows identify news effects

- MNAs are scheduled and occur at known times.
- Forecast surveys provide a pre-announcement expectation.
- The surprise component is realized news minus forecast news.
- High-frequency futures returns in short windows reduce confounding from other events.

Interpretation: The design isolates the asset-price reaction to the unexpected component of macro releases, not to the level of the announcement or to slow-moving macro conditions.

4.2 Why the output-gap result is informative

- Earlier literature often compares recessions with expansions.
- This paper shows that the strongest variation occurs within expansions.
- Sensitivity is highest early in expansions when output is below trend.
- Sensitivity becomes near zero late in expansions when output is close to or above trend.

Interpretation: Because peak sensitivity does not coincide with recessions or the highest VIX periods, the result is difficult to explain purely with risk or volatility. It points toward macro-policy interpretation.

4.3 Ruling out alternative measurement stories

The paper checks several alternatives:

- different announcement windows;
- broader versus narrower sets of MNAs;
- FOMC-day removal;
- individual announcement regressions;
- good-news versus bad-news decompositions;
- volatility, VIX, recession, and price-dividend controls;
- MMS forecast data as an alternative to Bloomberg forecasts.

Interpretation: These checks are meant to show that changing sensitivity is not due to changing surprise measurement, market microstructure, different announcement composition, or a pre-announcement drift effect.

4.4 Why bond and survey evidence are central

- Stock responses alone cannot separate cash-flow news from discount-rate news.
- Bonds react directly to expected future risk-free rates.
- If good news causes expected rate hikes, bonds should fall around announcements.
- Survey measures of perceived policy responsiveness provide independent evidence about how investors view the Fed's reaction function.

Interpretation: The paper uses comovement between stocks and bonds to infer the component of macro news that comes from expected monetary policy. Survey evidence validates that interpretation.

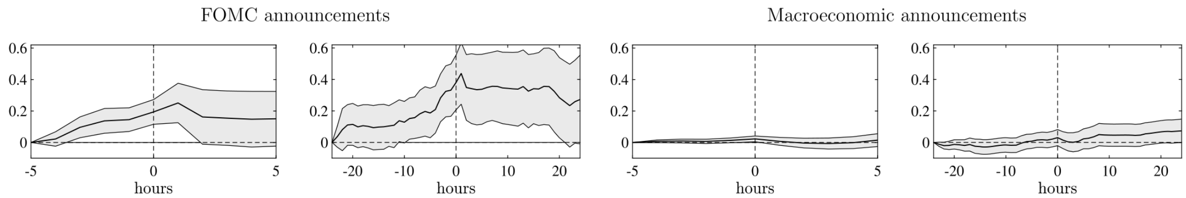
5 Tables and figures

5.1 Figure 1: cumulative returns around announcements

Read:

- The first two panels show FOMC announcement days, where stock returns display the well-known pre-announcement drift.
- The right panels show non-FOMC macroeconomic announcements.
- For MNAs, there is weak evidence of unconditional pre-announcement drift.
- This makes MNAs useful for measuring conditional responses to surprise components rather than unconditional return drift.

Economic message: FOMC announcements and regular macro announcements differ mainly in first moments. MNAs do not have a strong unconditional pre-announcement drift, so the paper focuses on how stock returns respond to the surprise component of macro news.



5.2 Figure 2 and Figure 3: pre-Covid stock-market sensitivity and decompositions

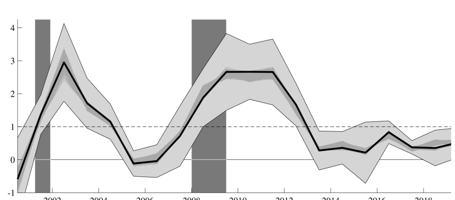
Read: Figure 2:

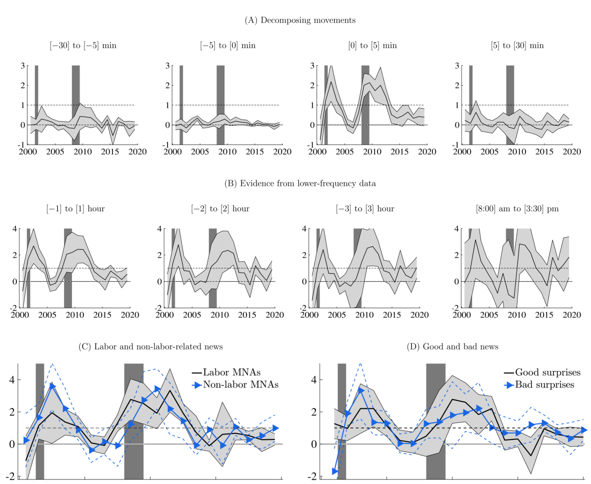
- Stock return sensitivity to macro news rises early in expansions.
- Sensitivity is lowest late in expansions, even though the economy is still growing.
- The sensitivity pattern is countercyclical with respect to the level of output relative to potential output.
- The benchmark news factor is built from the four most influential announcements: nonfarm payrolls, initial jobless claims, ISM manufacturing, and consumer confidence.

Read: Figure 3:

- The time-varying sensitivity is robust across announcement-window choices.
- Removing FOMC days does not eliminate the cyclicality.
- Labor-related and non-labor-related announcements display similar sensitivity patterns.
- Good-news and bad-news sensitivities are broadly similar, so the result is not driven only by one side of the news distribution.

Economic message: The stock-market response to macro news varies over the business cycle even outside recessions. The result survives many decompositions, which supports the view that the sensitivity series captures a genuine macro-finance state variable.





5.3 Figure 4 and Figure 5: Covid and post-Covid extensions

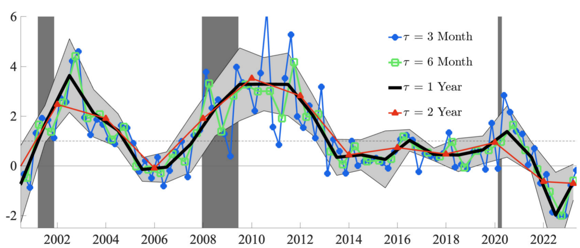
Read: Figure 4:

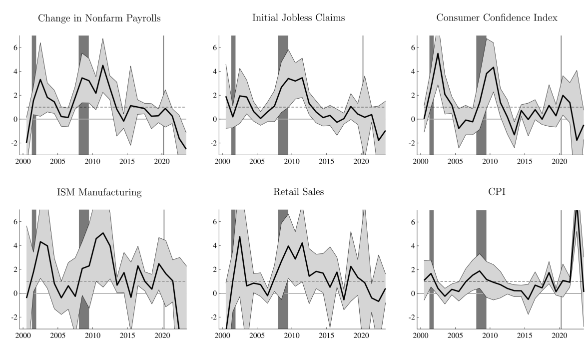
- Extending the sample through 2023 preserves the cyclical pattern documented before Covid.
- Stock return sensitivity rises sharply after the 2020 recession.
- In 2022–2023, the sensitivity turns negative: positive macro surprises can lower stock prices.
- This coincides with a period of high inflation and aggressive expected tightening.

Read: Figure 5:

- Individual announcement regressions show that the cyclical pattern is not specific to only one macro release.
- The first five individual macro surprises replicate the main business-cycle sensitivity pattern.
- CPI behaves differently: pre-Covid inflation surprises do not drive the main stock response, but post-Covid inflation becomes more important.

Economic message: After Covid, the Fed reaction function perceived by markets changes. Output-gap information still matters, but inflation becomes a dominant reason why good macro news can be bad news for stocks.





5.4 Tables 1 and 2: economic drivers of stock return sensitivity

Read: Table 1:

- In the pre-Covid sample, the output gap has a negative and significant coefficient in many specifications.
- A more negative output gap is associated with larger stock sensitivity to macro news.
- Changes in one-year Treasury bills are also strongly negative in the stock-sensitivity regression.
- Term premia are positive and significant in multivariate specifications.
- VIX, recession measures, and price-dividend ratios do not overturn the central output-gap result.

Read: Table 2:

- Including the post-Covid sample preserves the importance of output gap and expected interest-rate changes.
- Specifications that allow coefficients to differ after 2020 show that inflation becomes a strong driver of sensitivity.
- Post-Covid, positive inflation surprises and monetary-tightening expectations help explain negative stock responses to good macro news.

Economic message: Before Covid, output gap and interest-rate expectations explain the cyclical sensitivity of stocks to macro news. After Covid, inflation becomes a major input into perceived policy tightening.

Table 1
Economic drivers behind cyclical return responses: Excluding post-Covid periods.

	β_0	Output gap	ΔT-bill	T-bill	Inflation	Term premia	PD ratio	Recession prob.	VIX	R^2
(1)	0.96***	-0.35***								0.13
	(0.10)	(0.06)								
(2)	0.93***		-2.72***							0.12
	(0.11)		(0.66)							
(3)	0.96***			-0.31***						0.11
	(0.10)			(0.06)						
(4)	0.88***				-0.30**					0.10
	(0.11)				(0.12)					
(6)	0.82***					0.62***				0.12
	(0.10)					(0.10)				
(7)	0.92***						-2.15***			0.11
	(0.11)						(0.61)			
(8)	0.99***							0.03***		0.12
	(0.10)							(0.01)		
(9)	0.97***								0.06***	0.11
	(0.11)								(0.02)	
(10)	0.94***	-0.33***	-2.21***							0.14
	(0.10)	(0.06)	(0.64)							
(11)	0.95***	-0.34***	-2.27***		0.05					0.14
	(0.10)	(0.06)	(0.65)		(0.12)					
(12)	0.95***	-0.32**	-2.29***	-0.01	0.05					0.14
	(0.10)	(0.14)	(0.71)	(0.13)	(0.12)					
(13)	0.83***	-0.27***	-2.16***		0.01	0.47***				0.16
	(0.09)	(0.06)	(0.61)		(0.12)	(0.09)				
(14)	0.94***	-0.29***	-2.42***		0.05		-0.59			0.14
	(0.10)	(0.08)	(0.74)		(0.12)		(0.79)			
(15)	0.85***	-0.25***	-2.13***		0.03	0.45***		0.01	-0.01	0.16
	(0.10)	(0.06)	(0.69)		(0.12)	(0.11)		(0.01)	(0.02)	

Notes: We only report the coefficients associated with β^τ in the regression under both univariate and multivariate specifications. Output gap is the difference between real and potential GDP. Inflation is log changes in the GDP deflator; T-bill (3m) is the 3-month Treasury bill rate; and ΔT-bill (3m) is the difference between the one-quarter-ahead forecast and nowcast rate of the 3-month Treasury bill. The Treasury term premium for a maturity of 10 years is constructed by Adrian et al. (2013). The PD ratio is the price-to-dividend ratio from Shiller, and the VIX is the CBOE volatility index. The recession probability is the nowcast recession probability. All nowcast and forecast variables are collected from the Survey of Professional Forecasters. All variables are demeaned. Non-forecasted predictor variables are lagged one quarter. We use the four baseline macroeconomic announcements. We report Newey-West-adjusted standard errors. The estimation sample period is from 1999 to 2019. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Table 2
Economic drivers behind cyclical return responses: Including post-Covid periods.

		Output gap	ΔT-bill	Inflation	Term premium		Value
Specification:	$\beta^\tau = \beta_0 + \beta_1' Z_{\tau-1}$						
(1)	β_1	-0.25***	-2.44***	-0.05	0.45***	β_0	0.82***
		(0.05)	(0.56)	(0.06)	(0.08)		(0.09)
						R^2	0.14
(2)	β_1	-0.26***	-2.60***		0.46***	β_0	0.82***
		(0.05)	(0.54)		(0.08)		(0.09)
						R^2	0.14
Specification:	$\beta^\tau = \beta_0 + \beta_1' \mathbb{1}_{\tau<2020} Z_{\tau-1} + \beta_2' \mathbb{1}_{\tau\geq 2020} Z_{\tau-1}$						
(3)	β_1	-0.27***	-2.15***	0.01	0.47***	β_0	0.83***
		(0.06)	(0.61)	(0.12)	(0.09)		(0.09)
	β_2	-0.07	-3.43***	-0.17	0.15	R^2	0.14
		(0.10)	(1.33)	(0.12)	(0.26)		
(4)	β_1	-0.27***	-2.15***		0.47***	β_0	0.82***
		(0.06)	(0.59)		(0.09)		(0.08)
	β_2		-3.16***	-0.24***		R^2	0.14
			(1.13)	(0.06)			
(5)	β_1	-0.27***	-2.17***		0.48***	β_0	0.78***
		(0.06)	(0.60)		(0.09)		(0.08)
	β_2		-5.54***			R^2	0.14
			(1.06)				

Notes: We only report the estimates associated with β^τ in the regression. Output gap is the difference between the output and its potential trend. Inflation is a GDP deflator; ΔT-bill is the difference between the one-quarter-ahead forecast and nowcast rate of the 3-month Treasury bill. The Treasury term premium of maturity 10 years is constructed by Adrian et al. (2013). All nowcast variables are collected from the Survey of Professional Forecasters. All variables are demeaned. These predictor variables are lagged one quarter. We use the benchmarked macroeconomic announcements. We report Newey-West-adjusted standard errors. The estimation sample period is from 1999 to 2022. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

5.5 Table 3 and Figure 6: bond-market and survey evidence

Read: Table 3:

- The stock coefficient is positive on average, while bond coefficients are negative.
- Eurodollar and Treasury futures respond negatively to the news factor, consistent with higher expected rates after good macro news.
- Output-gap and term-premium coefficients show that the bond response is strongest when the economy is closer to peak.
- Longer-maturity bonds have larger negative constants, indicating stronger average sensitivity to macro news.

Read: Figure 6:

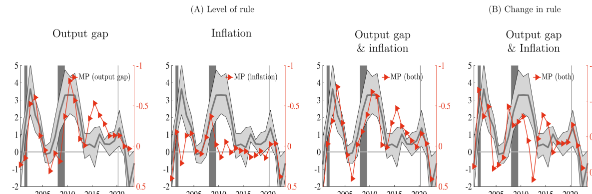
- The perceived interest-rate rule based on survey evidence moves closely with the stock sensitivity series.
- When perceived policy responsiveness is strong, macro news contains more discount-rate news.
- Annual changes in perceived interest-rate responsiveness line up with changes in stock-market sensitivity.

Economic message: Bond prices and surveys provide the paper's mechanism evidence. Stocks react most positively to macro news when investors expect the Fed to respond less aggressively; they react less or negatively when macro news implies rate hikes.

Table 3
Economic drivers behind cyclical return responses: Bond market evidence.

	Constant	Output gap	ΔT-bill	Term premium	R^2
Stock	0.82***	-0.26***	-2.60***	0.46***	0.14
	(0.09)	(0.05)	(0.54)	(0.08)	
EuroDollar 3m	-0.05***	-0.01***	0.06	-0.02***	0.14
	(0.01)	(0.00)	(0.05)	(0.00)	
EuroDollar 12m	-0.14***	-0.02***	-0.09	-0.07***	0.24
	(0.01)	(0.01)	(0.08)	(0.01)	
Treasury 5y	-0.61***	-0.03*	-0.30	-0.25***	0.25
	(0.04)	(0.02)	(0.26)	(0.04)	
Treasury 10y	-0.86***	-0.02	-0.49	-0.32***	0.24
	(0.05)	(0.02)	(0.34)	(0.05)	

Notes: We only report the estimates associated with β_0 (=constant) and β_1 in $\beta^\tau = \beta_0 + \beta_1' Z_{\tau-1}$ in the regression under multivariate specifications. We consider future returns defined over 30-minute intervals around MNAs for the E-mini stock, EuroDollar of maturity up to 3- and 12-months, Treasury of maturities 5- and 10-years, respectively. Output gap is the difference between the output and its potential trend. ΔT-bill is the difference between the one-quarter-ahead forecast and nowcast rate of the 3-month Treasury bill. The Treasury term premium of maturity 10 years is constructed by Adrian et al. (2013). All variables are demeaned. These predictor variables are lagged one quarter. We use the benchmarked macroeconomic announcements. We report Newey-West-adjusted standard errors. The estimation sample period is from 1999 to 2022. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.



5.6 Equation panels: surprise construction and sensitivity regressions

Read:

- The first panel shows how MNA surprises are standardized.
- Equation (1) is the main high-frequency stock-return regression.
- Equation (2) compresses announcements into the estimated scalar macro-news factor.
- Equation (3) separates good-news and bad-news responses.
- Equation (4) links time-varying sensitivity to macro and financial state variables.

Economic message: These equations summarize the empirical design. The paper first estimates how stocks respond to a macro-news factor, then explains the time variation in that response using state variables related to Fed reaction expectations.

Individual MNA surprises (after
vector X_t, whose ith component is

$$X_{i,t} = \frac{\mathrm{MNA}_{i,t} - E_{t-\Delta}(\mathrm{MNA}_{i,t})}{\mathrm{Normalization}}.$$

$$r_{t-\Delta_l}^{t+\Delta_h} = \alpha^\tau + \beta^\tau \gamma' X_t + \epsilon_t, \tag{1}$$

where the vector X_t contains various MNA surprises; γ measures the

$$r_{t-\Delta_l}^{t+\Delta_h} = \alpha^\tau + \beta^\tau \hat{X}_t + \epsilon_t, \tag{2}$$

regression.

$$r_{t-\Delta}^{t+\Delta} = \alpha^\tau + \beta^\tau_{\mathrm{good}} \gamma' X_{\mathrm{good},t} + \beta^\tau_{\mathrm{bad}} \gamma' X_{\mathrm{bad},t} + \epsilon_t. \tag{3}$$

$$r_{t-\Delta}^{t+\Delta} = \alpha^\tau + \beta^\tau \gamma' X_t + \epsilon_t, \quad \beta^\tau = \beta_0 + \beta_1' Z_{\tau-1}. \tag{4}$$

6 Section-by-section notes

6.1 Section 1: Introduction

Purpose of the section. The introduction explains a puzzle in macro-finance: the same macroeconomic news can move stocks in opposite directions at different times. Strong employment, production, or spending news can be bullish when investors focus on stronger cash flows, but bearish when investors infer that the Federal Reserve will raise rates.

Main conceptual decomposition.

- A positive macro surprise contains **cash-flow news**: firms may sell more, employment may be stronger, and aggregate demand may raise expected earnings.
- The same positive surprise can contain **risk-free-rate news**: if the Fed is expected to tighten, future short rates rise.
- Higher future short rates reduce equity valuations through discount rates.
- The net stock-market response is therefore the cash-flow effect minus the monetary-policy/discount-rate offset.

Core empirical claim. The stock-market sensitivity to macro news is time-varying. It is largest when output is below potential and smallest when output is near or above potential. This is especially striking because most of the variation happens within expansions rather than only across recession versus expansion states.

Why this matters. If stock sensitivity were only a recession phenomenon, one could attribute it to volatility, uncertainty, or risk premia. But because the sensitivity rises and falls inside expansions, the authors argue that it is better understood as variation in the perceived monetary-policy reaction function.

Post-Covid extension. The introduction also previews that the mechanism changes after 2020. In 2022–2023, high inflation makes good macro news more likely to be interpreted as bad news for stocks because investors expect stronger tightening. This shows that the relevant state variable is not always only slack/output gap; it is whatever input investors think the Fed is reacting to.

Interpretation: The introduction sets up the entire paper as a question about *interpretation*. The same announcement is not mechanically good or bad for stocks. Its meaning depends on the state-dependent policy rule investors have in mind.

6.2 Section 2.1: Data

Macroeconomic announcement data.

- The paper uses scheduled U.S. macroeconomic news announcements from Bloomberg.
- For each release, Bloomberg provides the realized data and survey expectations.
- The surprise is the realized release minus the pre-release forecast.
- The baseline period is January 1999 to December 2019; the extension goes through June 2023.
- The release set covers labor markets, production, confidence, consumer spending, inflation, GDP, housing, and other macro indicators.

Why scheduled announcements are useful. Scheduled MNAs occur at known times. Because the timing is predetermined and the forecast is observed, the high-frequency market movement around the release can be attributed to unexpected macro information rather than to the expected level of the data.

Standardization.

- Raw surprises have different units: payrolls are in jobs, CPI is in percentage points, ISM is an index, jobless claims are counts, and GDP is growth.
- The paper standardizes each surprise using a normalization term.
- The benchmark normalization uses cross-sectional forecast dispersion among individual forecasters.
- This means a one-unit standardized surprise is large relative to disagreement among forecasters.
- The paper also checks an unconditional time-series standard-deviation normalization.

High-frequency market data.

- Stock returns come from S&P 500 E-mini futures.
- Futures are used because many announcements occur before the cash stock market opens.
- Bond evidence uses Eurodollar and Treasury futures.
- Intraday prices come from TickData.
- Returns are constructed around short windows before and after the release time.

Macro-financial state variables. The explanatory state variables include output gap, changes in one-year Treasury bill rates, level of short rates, inflation, term premia, price-dividend ratio, VIX, and recession measures. These variables help test whether stock sensitivity is due to business-cycle slack, interest-rate expectations, risk appetite, valuation, or volatility.

Interpretation: Section 2.1 gives the identification ingredients: scheduled news, survey forecasts, high-frequency futures prices, and state variables. The goal is to measure the market price reaction to the unexpected component of macro news and then ask why that reaction varies over time.

6.3 Section 2.2: Estimating stock-return sensitivity to news before Covid

Step 1: Comparing MNAs to FOMC announcements.

- The authors first show that FOMC days exhibit pre-announcement drift.
- Regular MNAs do not show the same unconditional pre-announcement drift.
- This matters because the paper is about the conditional reaction to surprises rather than unconditional announcement premia.

Step 2: Estimating a macro-news factor. The paper estimates a vector of announcement weights, γ, so that $\gamma' X_t$ summarizes the macro-news content of all announcements. The factor is not a simple average; it gives larger weight to announcements that move markets more.

Step 3: Estimating time-varying sensitivity. The key coefficient is β_τ, the sensitivity of stock returns to the news factor in subperiod τ. A large β_τ means a given macro surprise produces a large stock return. A small or zero β_τ means the same news barely moves the market.

Pre-Covid finding.

- Sensitivity rises after recessions when the output gap is negative.
- It declines as expansions mature and output approaches potential.
- The largest change is within expansions, not only during NBER recessions.
- The sensitivity is not simply high when volatility is high.

Robustness and decomposition in Figure 3.

- Different return windows produce similar sensitivity dynamics.
- Removing FOMC days does not change the pattern.
- Labor and non-labor announcements show related dynamics.
- Good-news and bad-news responses move similarly, so the result is not only about negative surprises.
- Individual major announcements reproduce the main pattern.

Interpretation: Section 2.2 establishes the central empirical object of the paper: the stock-market macro-news sensitivity series. The important point is that the series is highly cyclical with slack, but not reducible to recession indicators or volatility.

6.4 Section 2.3: Stock-return sensitivity after the Covid pandemic

Why the extension matters. The post-Covid period is unusual because macroeconomic slack, inflation, and monetary policy move in ways not seen in the pre-Covid sample. This gives a stress test of the paper's interpretation.

Main post-Covid findings.

- The sensitivity rises sharply after the 2020 recession, consistent with the pre-Covid pattern of high sensitivity when the output gap is negative.
- During 2022–2023, sensitivity can turn negative.
- This means positive macro surprises can lead to negative stock returns.
- The reason is that investors interpret good news or high inflation as making monetary tightening more likely.

Role of CPI and inflation news. The individual-announcement analysis shows that inflation surprises become much more important after Covid. CPI surprises are not the main pre-Covid driver, but they matter strongly when inflation becomes the Fed's central concern.

How this strengthens the paper. The post-Covid result does not contradict the pre-Covid output-gap result. Instead, it generalizes it: stock sensitivity depends on the state variables that investors believe enter the Fed reaction function. Before Covid, slack/output gap is central; after Covid, inflation is central.

Interpretation: Section 2.3 shows that the mechanism is not mechanically tied to one variable. The common logic is perceived monetary-policy responsiveness: good macro news is bad for stocks when it raises expected rates enough to dominate cash-flow news.

6.5 Section 3.1: Stock-market evidence on economic drivers

Goal of this subsection. After estimating time-varying sensitivity, the paper asks what macro-financial states explain it. The key regression restricts sensitivity as

$$\beta_\tau = \beta_0 + \beta_1' Z_{\tau-1}.$$

This turns the sensitivity series into an object explained by output gap, rates, inflation, term premia, valuation, volatility, and recession variables.

Pre-Covid drivers in Table 1.

- The output gap is negative and significant in many specifications.
- A more negative output gap predicts higher stock sensitivity to macro news.
- Changes in short rates are also strongly related to sensitivity, consistent with expected policy tightening.
- Term premia matter in multivariate specifications.
- VIX, recession measures, and valuation measures do not eliminate the output-gap result.

Interpretation of the output-gap coefficient. A negative output gap coefficient means that when output is far below potential, stock responses to positive news are stronger. Economically, slack makes the Fed less likely to offset good news with immediate tightening, so cash-flow news dominates.

Post-Covid drivers in Table 2.

- Extending to 2023 preserves the relevance of the output gap.
- Allowing post-2020 coefficients reveals that inflation becomes very important.
- This matches the 2022–2023 experience: positive macro or inflation surprises raise rate-hike expectations.
- The post-Covid regressions therefore show an expanded policy-reaction channel rather than a breakdown of the model.

Interpretation: Section 3.1 provides the main explanatory regression evidence. It shows that the state-dependence in stock responses is tied to variables that plausibly govern policy expectations, not just to volatility or recession status.

6.6 Section 3.2: Bond-market evidence

Why bonds are necessary. Stocks combine cash-flow and discount-rate news. Bonds are more directly informative about the risk-free-rate component of macro news. If positive macro news raises expected short rates, bond prices should fall.

Bond instruments.

- Eurodollar futures capture expected short-term interest rates.
- Treasury futures capture rate expectations and term-structure effects at different maturities.
- The same macro-news factor is used so that stock and bond reactions are comparable.

Main bond results.

- Bond futures generally respond negatively to positive macro news.
- This is consistent with good news raising expected rates.
- Bond responses are stronger when the economy is closer to or above potential.
- Longer maturity instruments show stronger average sensitivity in some specifications.

Connection to the stock results. When the Fed is expected to tighten aggressively, the bond response becomes more negative and the stock response becomes weaker or negative. When the Fed is expected to respond mildly, bond-rate news is less offsetting and stock sensitivity is positive.

Interpretation: Section 3.2 is the mechanism test. It confirms that the discount-rate channel moves in the direction required to explain why good macro news sometimes fails to lift stocks.

6.7 Section 3.3: Survey forecasts and perceived policy rules

Why surveys are useful. Asset prices reveal how markets react, but they do not directly state what investors believe about the Fed's reaction function. Survey evidence gives an independent measure of perceived policy responsiveness.

Main survey logic.

- Forecasters' expected interest-rate responses to macro conditions are used to infer perceived policy rules.
- When the perceived rule is more aggressive, macro news has more rate news.
- When the perceived rule is less aggressive, macro news contains more unoffset cash-flow news.

Figure 6 evidence. The perceived interest-rate rule tracks the stock sensitivity series. Annual changes in perceived responsiveness line up with annual changes in stock sensitivity.

Interpretation: Section 3.3 completes the mechanism argument. It shows that the estimated sensitivity series is not just a statistical artifact; it lines up with independently measured beliefs about the Fed.

6.8 Section 4: Conclusion

Summary of the paper's answer. The stock market does not have a fixed reaction to macro news. The reaction depends on how much of the news investors interpret as cash-flow news versus future-rate news.

Pre-Covid conclusion. Before Covid, output-gap slack is the key state variable. When the economy is below trend, good news is good for stocks. When the economy is near peak, good news is more likely to imply Fed tightening and is less positive for stocks.

Post-Covid conclusion. After Covid, inflation becomes the dominant state variable because investors focus on how inflation affects the Fed. Good macro news or high inflation surprises can be bad for stocks when they imply higher rates.

Broader implication. The paper's broader contribution is to connect high-frequency macro announcement returns to a changing perceived monetary-policy rule. This helps explain why media narratives such as "good news is bad news" are not irrational or inconsistent; they reflect state-dependent discount-rate effects.

Interpretation: The conclusion reinforces the title: Wall Street reads Main Street through the Fed. Macro data matter for stocks because they reveal both fundamentals and policy reaction.

7 Expanded guide to every empirical exhibit

7.1 Figure 1: cumulative returns around announcements

What is being plotted. The figure plots cumulative S&P 500 E-mini returns around announcement windows. It contrasts FOMC announcements with regular macroeconomic news announcements.

Why it is included. The figure clarifies that FOMC announcements have a distinct pre-announcement drift, while regular MNAs do not have the same unconditional drift. This justifies focusing on surprise-conditioned MNA reactions rather than unconditional return patterns.

How to read it. The key visual question is whether returns move before the announcement even without conditioning on surprises. If they do, one must worry about pre-announcement drift. For MNAs, the unconditional drift is limited, so the main object becomes the high-frequency response to the surprise.

7.2 Figure 2: pre-Covid stock sensitivity

What is being plotted. Figure 2 plots estimated β_τ values for 1999–2019 and compares them to recession/expansion timing and macro state variables.

Key detail. The strongest sensitivity occurs after recessions and early in expansions. Sensitivity declines as expansions mature. This means the coefficient is not simply a recession dummy.

Economic reading. High sensitivity when slack is large means the market sees macro news as mostly cash-flow news. Low sensitivity near the peak means the market sees macro news as partially or fully offset by expected tightening.

7.3 Figure 3: decompositions and robustness

What is being tested. Figure 3 checks whether the sensitivity series changes when the authors vary the return window, remove FOMC days, split labor/non-labor releases, or separate good from bad surprises.

Why it matters. If the main result disappeared under one of these changes, it might reflect microstructure, an announcement subset, or asymmetry in bad news. The fact that it persists makes the evidence stronger.

7.4 Figure 4: sample through 2023

What changes after Covid. The basic business-cycle logic survives the 2020 recession, but 2022–2023 introduces negative sensitivity because inflation and tightening expectations dominate.

Why it matters. The negative coefficient is not a failure of the framework. It is a case where the Fed reaction function component overwhelms cash-flow news.

7.5 Figure 5: individual announcement regressions

Purpose. This figure checks whether one announcement drives the result. It shows that multiple announcements display the same pattern, while inflation releases become more important post-Covid.

Economic reading. The market's macro interpretation is broad-based. The changing role of CPI shows that the policy-reaction channel adapts to the macro regime.

7.6 Tables 1 and 2: stock-market drivers

Purpose. These tables formally test which state variables explain β_τ.

Main pre-Covid reading. Output gap and interest-rate changes matter. This supports the policy-reaction interpretation and weakens pure volatility/recession explanations.

Main post-Covid reading. Inflation becomes central. This is exactly what one should expect in a high-inflation regime in which the Fed is focused on inflation stabilization.

7.7 Table 3: bond-market evidence

Purpose. Table 3 asks whether macro news moves expected rates in the direction needed to explain the stock patterns.

Main reading. Good macro news tends to lower bond futures prices, consistent with higher expected rates. The response is especially informative when stocks respond weakly or negatively.

7.8 Figure 6: survey policy-rule evidence

Purpose. Figure 6 connects market-implied sensitivity to survey-implied perceived policy responsiveness.

Main reading. The survey evidence supports the claim that investors' perceived Fed reaction function changes over time and helps determine stock-market reactions to macro news.

8 Short summary

- The paper studies how stock prices react to macroeconomic news announcements from 1999 to 2023.
- The benchmark analysis focuses on high-frequency S&P 500 E-mini futures returns around scheduled announcements.
- Surprise components are measured using Bloomberg survey forecasts and normalized by forecaster disagreement.
- Stock sensitivity to macro news is countercyclical with respect to the output gap.
- The largest pre-Covid variation occurs within expansions, not simply in recessions.
- The sensitivity is high when the economy is below trend and low when the economy is close to peak.
- The pattern is robust to different announcement windows, different announcement sets, FOMC-day exclusions, and good/bad news splits.
- The output gap and expected interest-rate changes explain the pre-Covid sensitivity series.
- Bond futures respond in the opposite direction, implying that macro news changes expected future risk-free rates.
- Survey evidence supports the interpretation that investors perceive the Fed as more or less responsive depending on macro conditions.
- After Covid, inflation becomes a key determinant of stock sensitivity because inflation surprises raise expected tightening.
- The paper's core message is that stock markets read macro news through expected monetary policy.

9 Conclusion

9.1 Core conclusion

Elenev, Law, Song, and Yaron show that the stock market's reaction to macroeconomic news varies systematically over the business cycle. The response is strongest when output is below trend and weakest near the peak of expansions. The key reason is that macro news contains both cash-flow information and future-rate information, and the future-rate component depends on perceived Federal Reserve responsiveness.

9.2 Economic intuition

Positive macro news raises expected cash flows. But if investors think the Fed will respond aggressively, the same positive news also raises expected future risk-free rates. Higher expected rates reduce stock values. Therefore, the net stock response depends on how much the Fed is expected to offset the cash-flow effect. When slack is high, the Fed is expected to be less aggressive, so stocks rise strongly on good news. When the economy is tight or inflation is high, the Fed is expected to tighten, so good news can be muted or even negative for stocks.

9.3 Takeaway

The paper reframes macro announcement effects as a monetary-policy-expectations problem. Wall Street does not only read Main Street data directly. It reads Main Street through the Fed's perceived reaction function. This explains why the same macro surprise can be bullish in one state of the economy and bearish in another.